Netflix Customer Intelligence & AI Analytics Copilot

Complete Project Report

Portfolio Project • Data Science • Machine Learning • AI • MLOps

Project Repository

github.com/NandiniShilpkar/Netflix-Customer-Intelligence

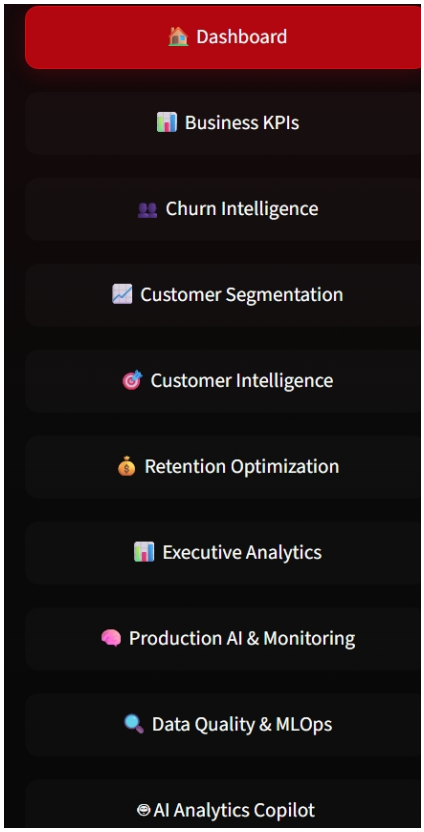
Prepared by Nandini Shilpkar

Version 1.0.0

A customer intelligence platform for churn prediction, customer 360, retention prioritization, business analytics, AI-assisted analysis, and monitoring.

Application Modules

The application contains **10 main modules**. The module navigation shown below is taken directly from the project application screenshot.



1. Dashboard	Overall customer intelligence overview and key application-level metrics.
2. Business KPIs	Business performance indicators for customers, revenue, engagement, and churn.
3. Churn Intelligence	Historical churn analysis, risk patterns, and churn-related insights.
4. Customer Segmentation	Behavior-oriented customer groups and segment-level churn analysis.
5. Customer Intelligence	Customer-level intelligence including Customer 360 and decision signals.
6. Retention Optimization	Retention targeting, recommended actions, and revenue-at-risk analysis.
7. Executive Analytics	Executive-level summaries, priority customers, and business impact.
8. Production AI & Monitoring	Monitoring of model, AI, data coverage, and production-readiness signals.
9. Data Quality & MLOps	Data quality checks, feature completeness, relationships, and MLOps reporting.
10. AI Analytics Copilot	Natural-language business analysis using structured analytics and local AI.

Module count: 10 — these are the main navigation modules presented in the application.

1. Executive Summary

Netflix Customer Intelligence & AI Analytics Copilot is a portfolio-grade analytics and machine-learning application designed to turn customer, subscription, viewing, payment, support, and feedback data into actionable retention intelligence. The system combines a historical churn model with business rules and customer-level analytics to prioritize customers for retention actions.

The application provides executive analytics, churn intelligence, customer segmentation, customer 360 profiles, retention optimization, an AI Analytics Copilot, data-quality/MLOps checks, and production AI/model monitoring. A local LLM (Ollama, llama3.2:3b) is used for business-question answering when the external OpenAI API is not available.

2. Business Problem

A streaming business needs to identify customers who may churn, understand why they are at risk, estimate revenue exposure, and decide where retention teams should focus first. The project addresses this by connecting customer behavior and operational signals to machine-learning risk scores and business decisions.

3. Project Objectives

• Build a reliable customer analytics data foundation.
• Clean and validate customer and behavioral data.
• Engineer customer-level features for churn prediction.
• Train and validate a churn model using historical snapshots.
• Translate risk scores into business actions and revenue-at-risk estimates.
• Provide a Customer 360 view for individual customers.
• Provide an AI Analytics Copilot for natural-language business questions.
• Monitor data quality, prediction coverage, model availability, and AI service health.

4. Data Architecture & Source Data

The application works with nine core source tables. The source-data inventory visible in the supplied screenshots confirms the following row counts:

Table	Rows
customers	8,000
subscriptions	8,875
subscription_plans	3
viewing_activity	66,422
content	500
payments	110,735
support_tickets	6,120
customer_feedback	5,046
churn_labels	8,000

5. Data Engineering & Data Quality

The project includes inspection, cleaning, and relationship validation. Duplicate customer records were removed during cleaning, and relationship validation reported zero unmatched relationships. The live dashboard distinguishes between a quality-check score and raw feature completeness: the defined validation checks can pass even while individual feature values remain missing.

Key observed quality metrics

Metric	Value
Quality checks / score	100%
Missing feature values	10,545
Missing feature rate	6.74%
Feature completeness	68%
Customer coverage	100%
Duplicate feature rows	0

Important interpretation: the 100% quality score shown by the application represents the configured validation checks passing. It does not mean that every raw feature is complete; the dashboard explicitly reports 10,545 missing feature values and 68% feature completeness.

6. Machine Learning — Churn Prediction

The active model is a historical Random Forest churn classifier. It was trained on customer-level historical snapshots with the target churn_next_30_days. The active artifact uses 300 trees, min_samples_leaf=20, balanced class weights, and random_state=42.

Training data: 32,709 rows; 1,109 positive churn cases and 31,600 negative cases. Historical snapshots covered January–May 2026.

Rolling validation

Validation	ROC-AUC	Average Precision	F1
Feb → Mar	0.610	0.050	0.101
Feb+Mar → Apr	0.598	0.058	0.110
Feb+Mar+Apr → May	0.630	0.075	0.130

The validation results indicate moderate predictive performance. The project therefore positions the model as decision support rather than a guaranteed production churn oracle.

7. Customer Decision Engine & Retention Optimization

The current scored cohort contains 6,260 customers. Business decisioning translates risk and customer value into operational priorities.

Decision	Customers
Monitor	3,216
Engage	1,833
Low Priority	1,154

Save Now	57

Revenue at risk: approximately $4,517.72 monthly and $54,212.64 annualized. The retention optimization view identifies 1,890 targeted customers and models a 30% annual recovery scenario of approximately $16,263.79.

8. Customer 360

Customer 360 consolidates customer profile information, plan, price, churn risk, health score, value score, priority score, decision, recommended action, revenue-at-risk metrics, viewing behavior, support activity, and feedback signals into one customer-level profile.

Example shown in the project: customer CUST000685 has a churn risk of approximately 72.9%, health score 53.3, decision 'Save Now', Premium plan, $22.99 monthly price, and $275.88 annualized revenue at risk.

9. AI Analytics Copilot

The AI Analytics Copilot provides a natural-language interface for customer, churn, subscription, viewing, and business-performance questions. Customer 360 queries can be routed to structured customer data. The local AI service uses Ollama with llama3.2:3b, which keeps the project usable without depending on a paid external LLM API.

The Copilot is designed to ground responses in available analytics and avoid inventing customer IDs or unsupported numbers.

10. Production AI & Model Monitoring

The monitoring layer provides a production-readiness view covering model availability, prediction coverage, feature quality, duplicate rows, AI service availability, and churn-risk distribution.

Monitoring metric	Current value
Customers scored	6,260
Low-risk customers	2,237
Medium-risk customers	3,314
High-risk customers	709
Prediction coverage	100%
Production readiness	83.33%
Duplicate feature rows	0
Missing feature values	10,545

The supplied application screenshots show the churn model and local AI as healthy, while feature quality is flagged for review because missing values remain. This is an important production-readiness signal rather than a defect hidden by the dashboard.

11. Technology Stack

Application: Python, Streamlit
Data: Pandas, NumPy, PostgreSQL
ML: Scikit-learn, Random Forest, Joblib
Visualization: Plotly / Streamlit charts
AI: Ollama + llama3.2:3b; optional OpenAI integration
Engineering: SQLAlchemy, psycopg, python-dotenv
Development: GitHub Codespaces, Git/GitHub

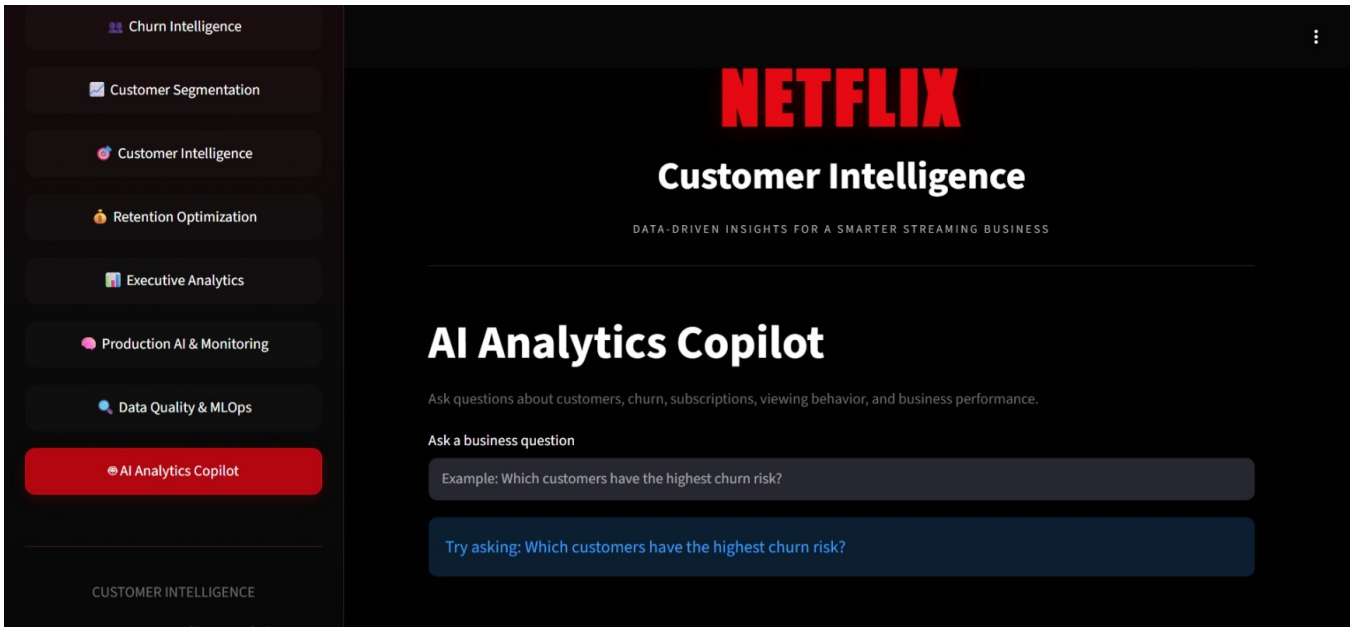
12. Application Modules

- Executive Analytics
- Churn Intelligence
- Customer Segmentation
- Customer Intelligence / Customer 360
- Retention Optimization
- Production AI & Model Monitoring
- Data Quality & MLOps
- AI Analytics Copilot

13. Project Screenshots

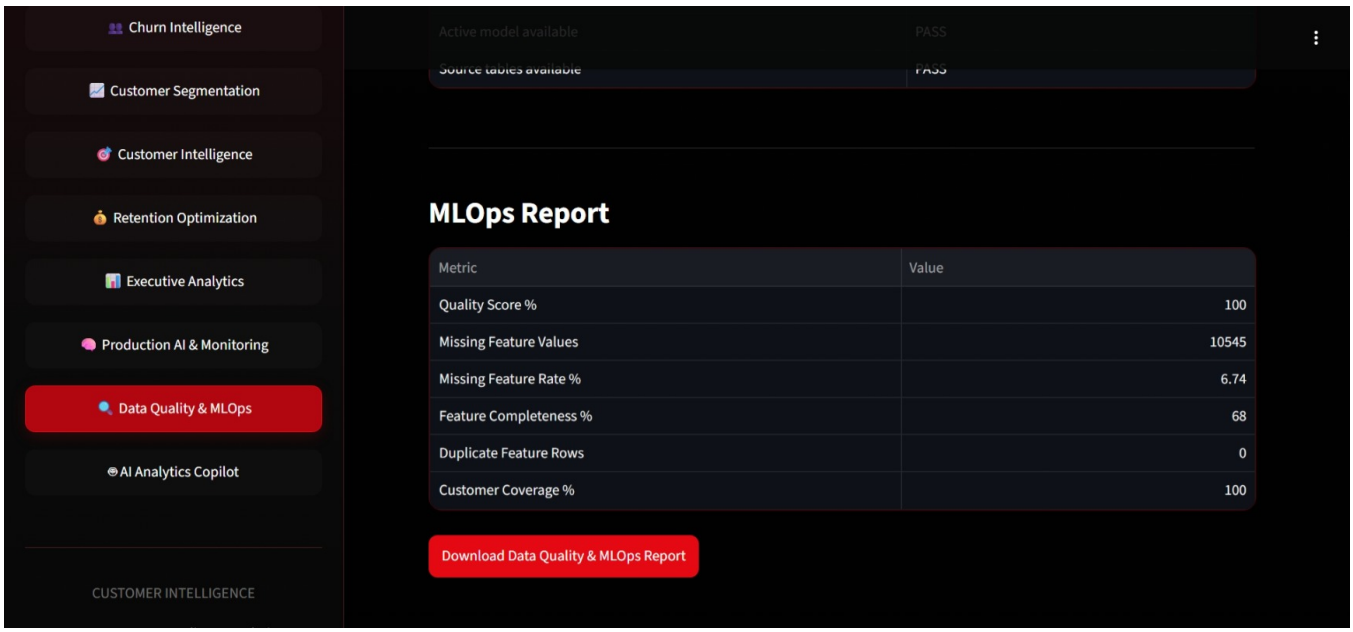
The following 15 screenshots were supplied with the project request and are included as visual evidence of the live application. Captions describe the visible module/functionality; the images themselves are treated as the source of truth for the UI.

Screenshot 1



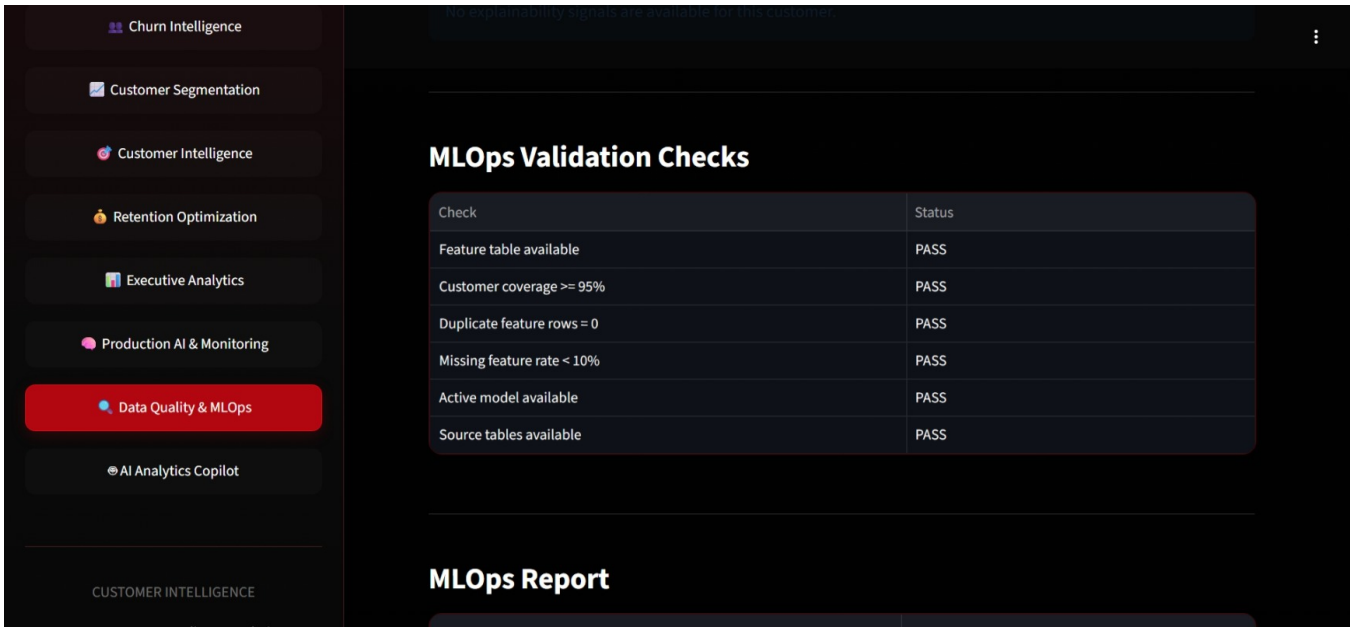
Live application — AI Analytics Copilot

Screenshot 2



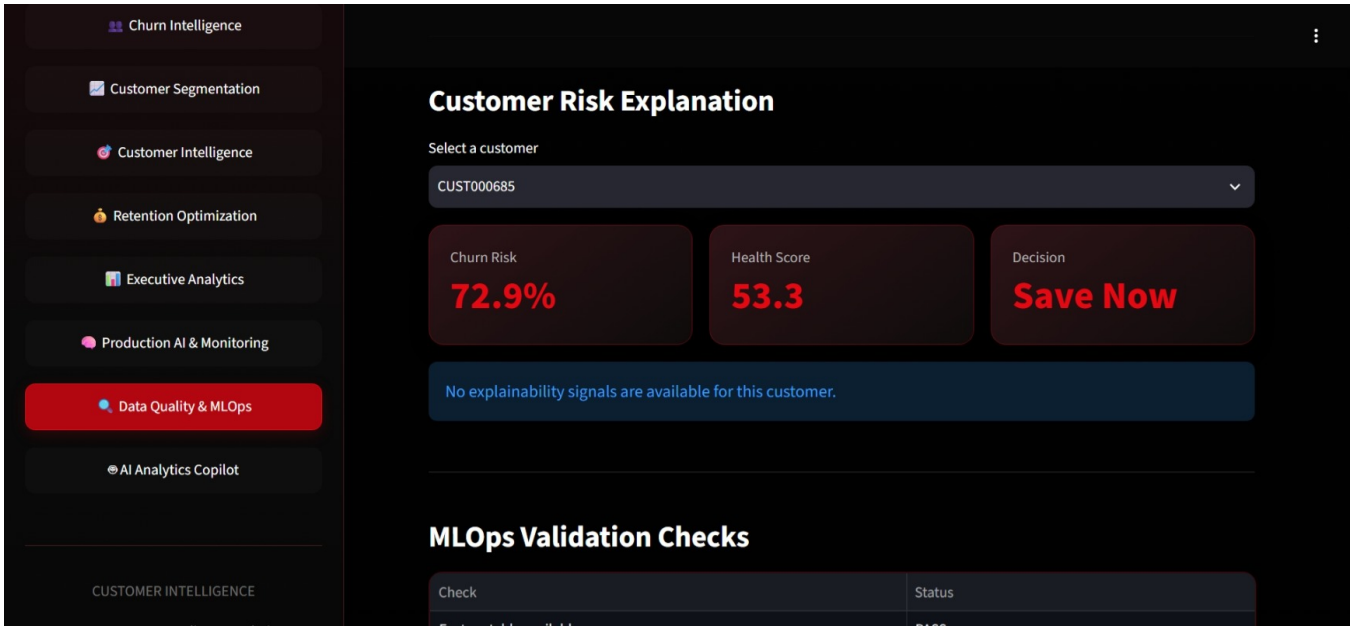
Data Quality & MLOps — validation checks and report

Screenshot 3



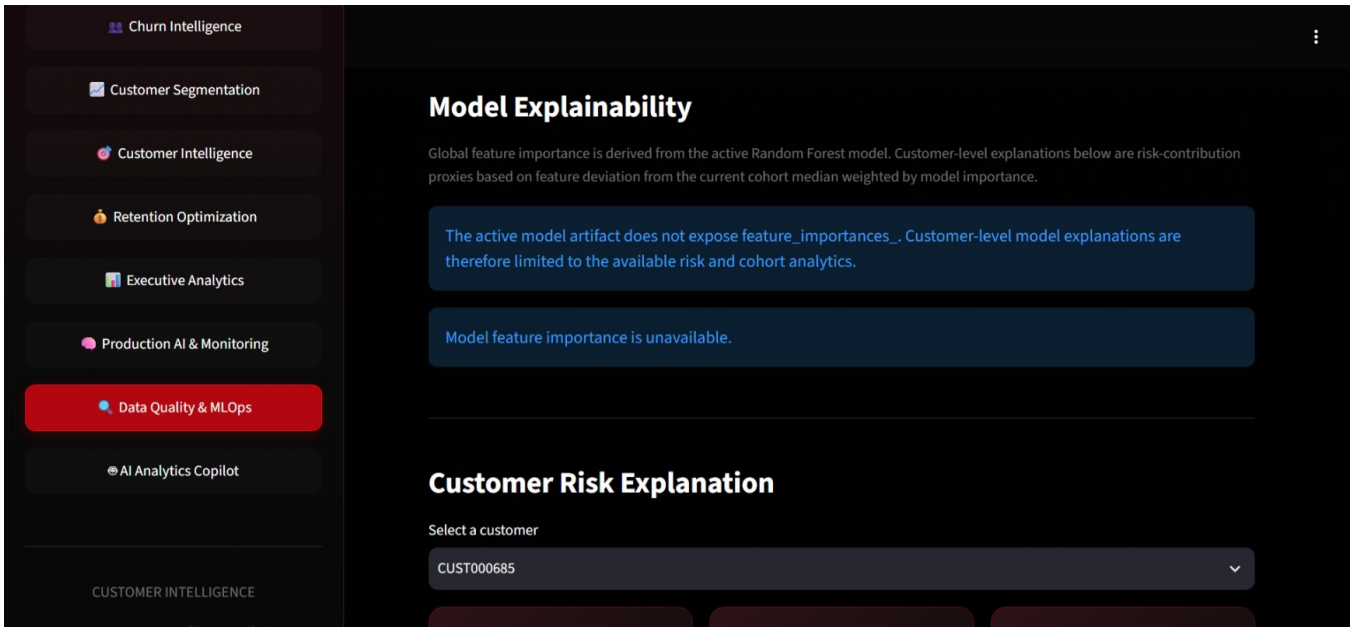
Data Quality & MLOps — customer risk explanation

Screenshot 4



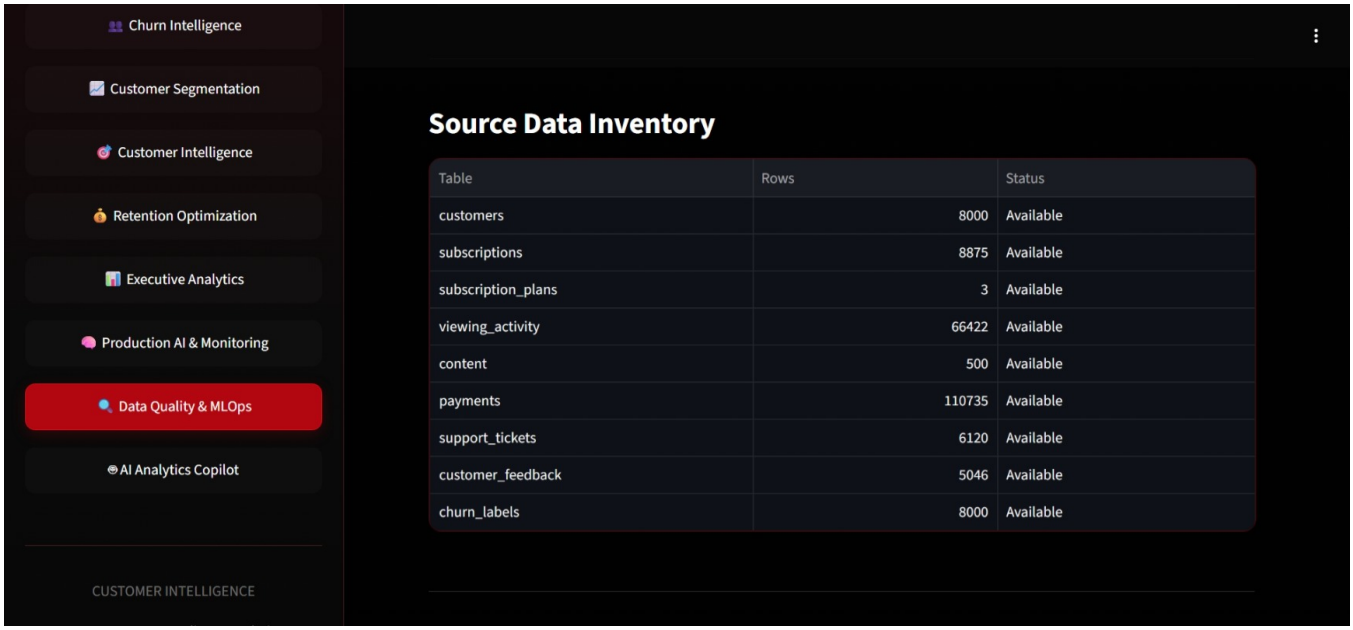
Model Explainability and customer risk explanation

Screenshot 5



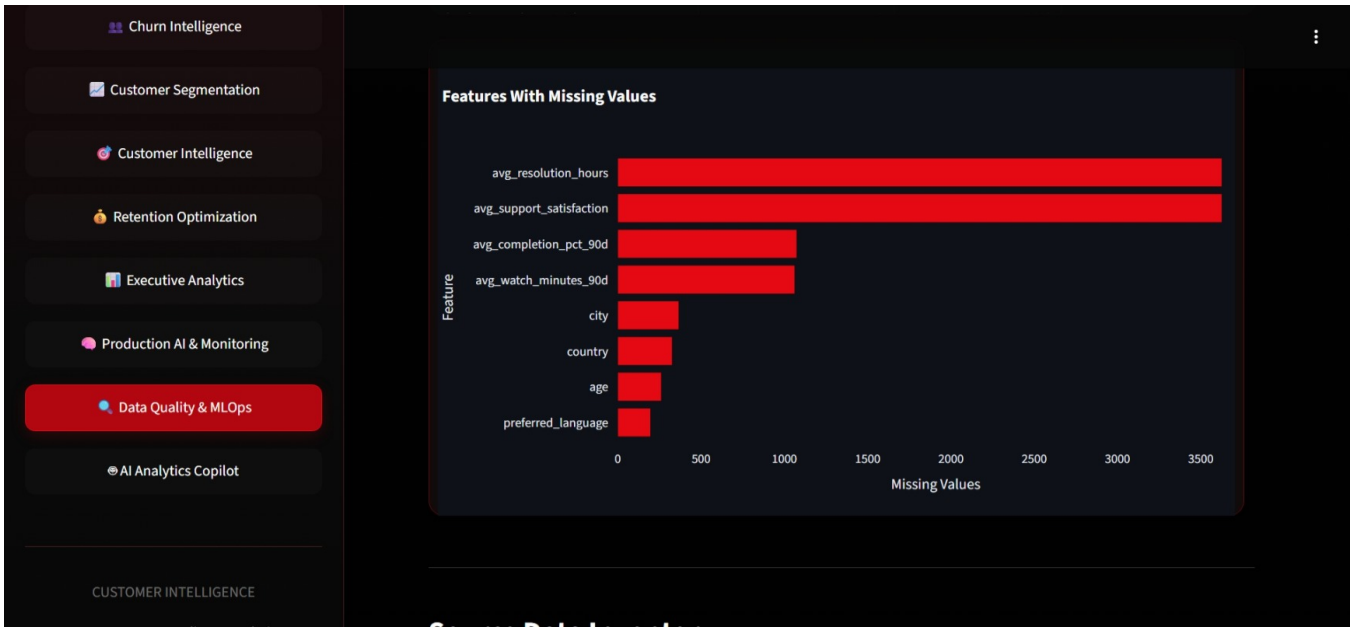
Source Data Inventory

Screenshot 6



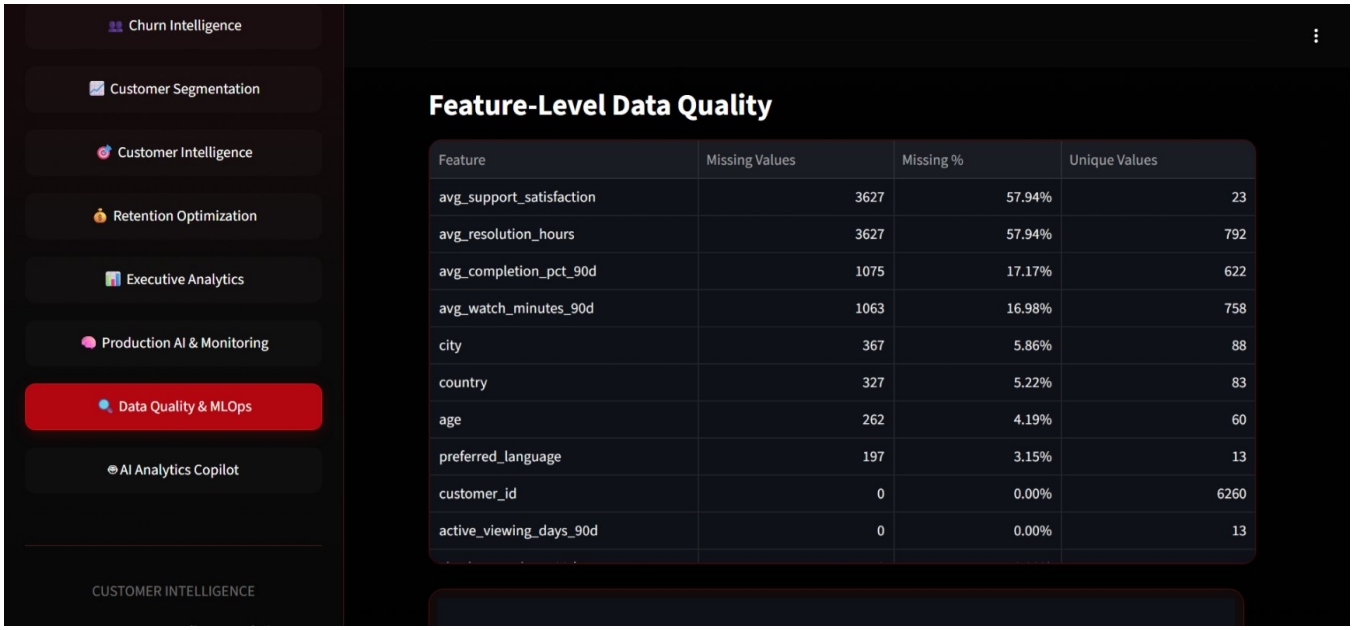
Missing-feature distribution

Screenshot 7



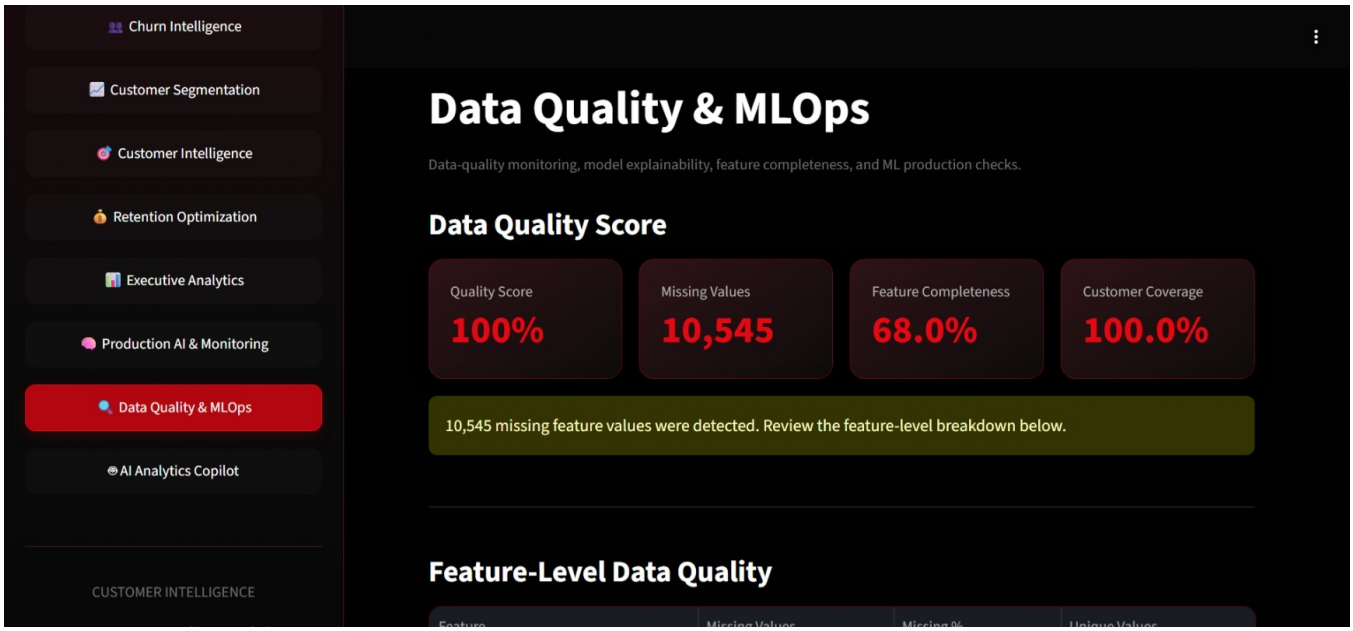
Feature-level data quality table

Screenshot 8



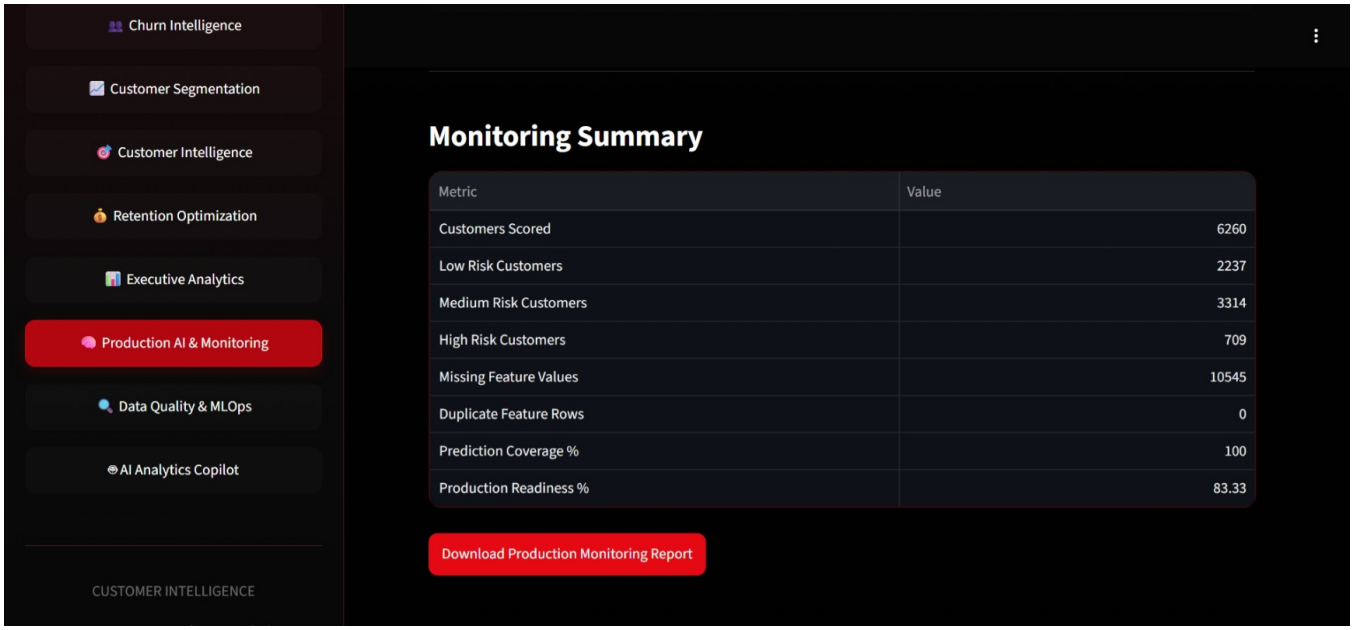
Data Quality & MLOps overview

Screenshot 9



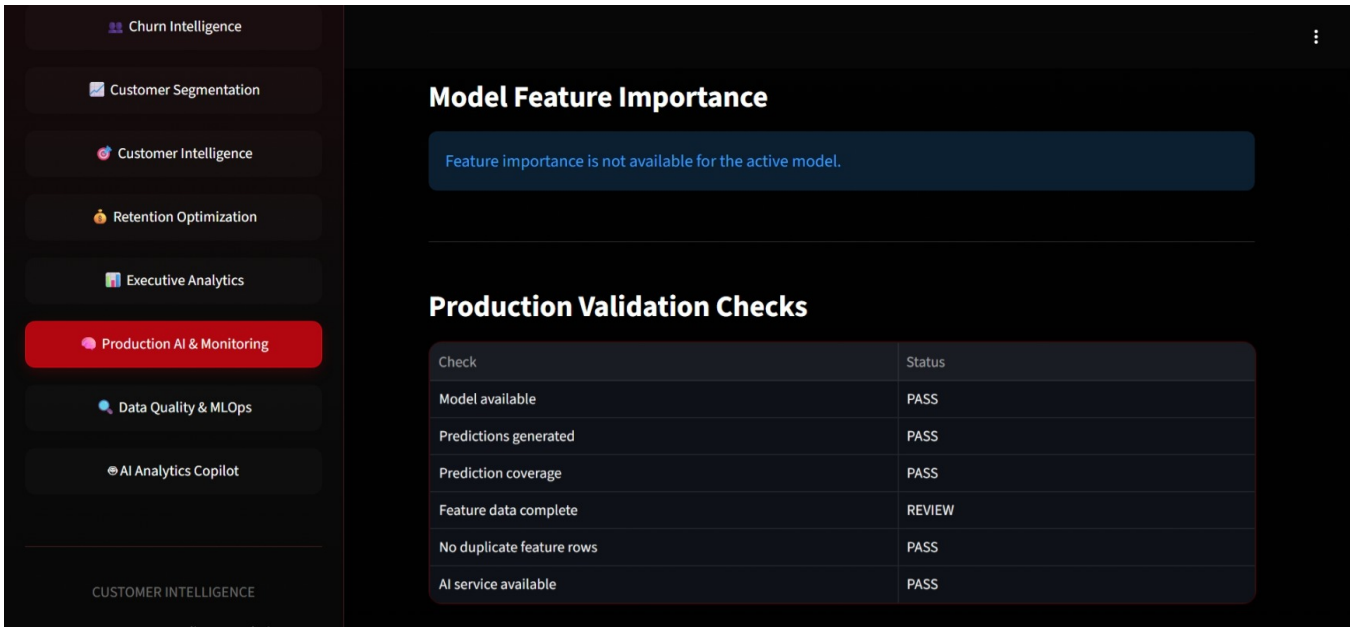
Production monitoring summary

Screenshot 10



Model feature importance and validation area

Screenshot 11



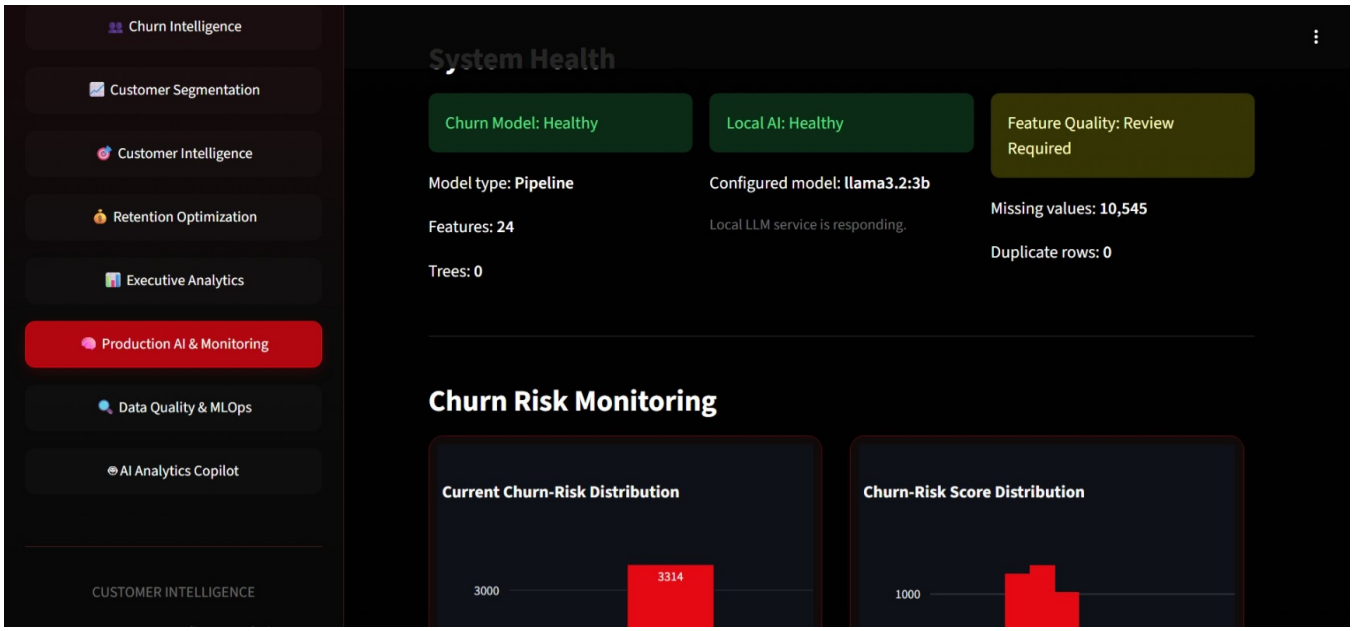
Churn-risk monitoring

Screenshot 12



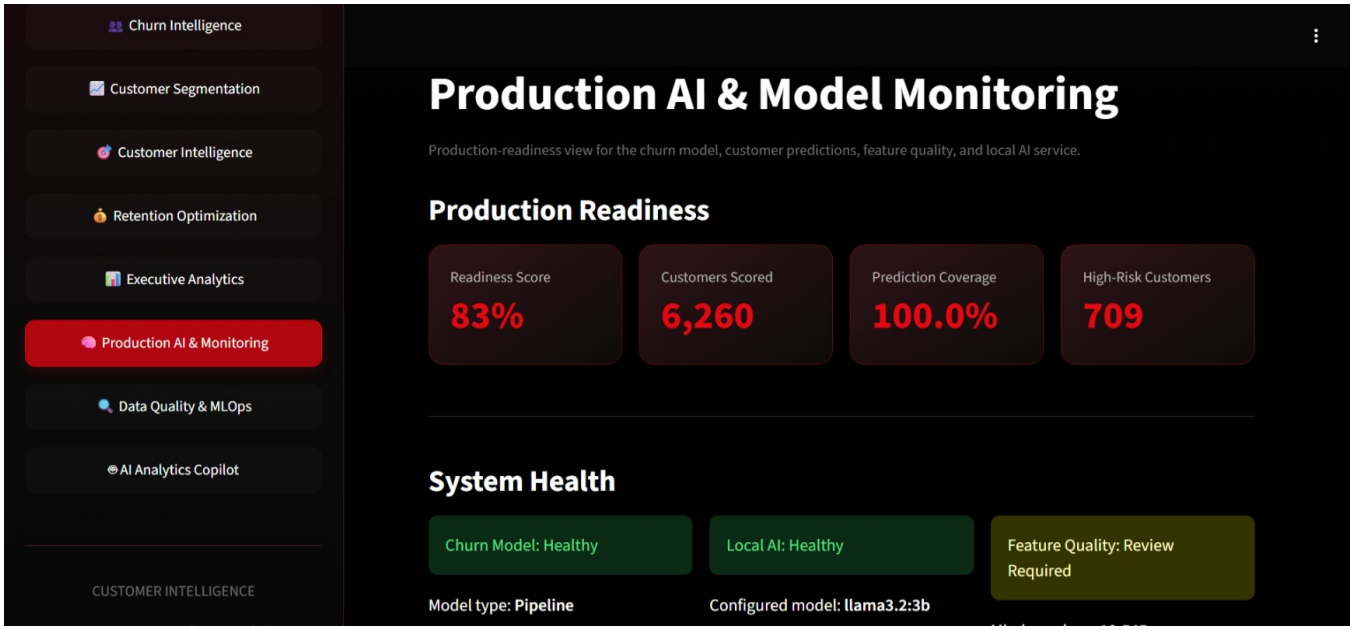
Production system health and churn-risk monitoring

Screenshot 13



Production AI & Model Monitoring overview

Screenshot 14



Additional production monitoring view

Screenshot 15



Additional project application view

14. Business Impact & Portfolio Value

The project demonstrates an end-to-end workflow rather than a standalone ML notebook. It connects raw operational data to cleaning, feature engineering, predictive modeling, business prioritization, customer-level analytics, AI assistance, and production monitoring.

For a Data Scientist / ML Engineer / AI-ML portfolio, the strongest aspects are the combination of supervised learning, business decisioning, customer 360 analytics, revenue-at-risk estimation, natural-language analytics, and MLOps-style validation.

15. Limitations & Responsible Interpretation

- The churn model has moderate validation performance and should be treated as decision support.
- Feature completeness is 68%, with 10,545 missing feature values visible in the monitoring layer.
- Customer-level explainability is limited by the active model artifact and available feature-importance signals.
- Revenue recovery values are scenario estimates, not guaranteed realized revenue.
- The live GitHub Codespaces URL may depend on the Codespace being active and publicly accessible.

16. Future Enhancements

- Improve feature completeness and missing-data handling.
- Retrain and compare stronger churn models such as calibrated gradient boosting models.
- Add SHAP-based global and customer-level explanations.
- Add model drift, data drift, and performance monitoring over time.
- Add automated retraining and model registry/versioning.
- Add role-based access control and production deployment infrastructure.
- Add controlled A/B testing for retention interventions.

17. Project Links

Live Application: https://netflix-customer-intelligence-tsmnzjnygbsxbgtbfbvcnd.streamlit.app/
GitHub Repository: https://github.com/NandiniShilpkar/Netflix-Customer-Intelligence

18. Conclusion

Netflix Customer Intelligence & AI Analytics Copilot demonstrates how customer data, machine learning, business analytics, and AI can be combined into a practical decision-support platform. The project is structured to show not only predictive modeling, but also the steps required to make model outputs useful to business teams: prioritization, customer context, revenue exposure, natural-language access, and production monitoring.

Prepared as a project showcase report.
All quantitative values in this report are based on the project results and screenshots supplied in the conversation.